

1015. Smallest Integer Divisible by K

Difficulty: Medium

Link: <https://leetcode.com/problems/smallest-integer-divisible-by-k/description/?envType=daily-question&envId=2025-11-25>

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Problem:

Given a positive integer k , you need to find the **length** of the **smallest** positive integer n such that n is divisible by k , and n only contains the digit 1 .

Return the **length** of n . If there is no such n , return -1.

Note: n may not fit in a 64-bit signed integer.

Example 1:

Input: $k = 1$

Output: 1

Explanation: The smallest answer is $n = 1$, which has length 1.

Example 2:

Input: $k = 2$

Output: -1

Explanation: There is no such positive integer n divisible by 2.

Example 3:

Input: $k = 3$

Output: 3

Explanation: The smallest answer is $n = 111$, which has length 3.

Constraints:

- $1 \leq k \leq 10^5$